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Article in *The Journal of Strength and Conditioning Research* · October 2023

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Shoulder and Scapular Muscle Activity During Low and High Plank Variations With Different Body-Weight-Bearing Statuses

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Abstract

Can, EN, Harput, G, and Turgut, E. Shoulder and scapular muscle activity during low and high plank variations with different body-weight-bearing statuses. *J Strength Cond Res* XX(X): 000–000, 2023—This study investigated the activation levels and activation ratios of shoulder and scapular muscles during low plank (LP) and high plank (HP) variations, with varying levels of body-mass support, including 4-point, 3-pod, and 2-point body-weight-bearing (BWB) statuses. The study was conducted with 21 healthy men (mean $\pm$ SD, 26 $\pm$ 6.5 years of age and 24.4 $\pm$ 2.4 kg·m⁻² BMI). Ten different plank exercises were performed in a mixed order by changing elbow joint positions (LP and HP) and BWB statuses (2-point, 3-pod, and 4-point). Activation levels of the lower trapezius (LT), middle trapezius (MT), upper trapezius (UT), biceps brachii, triceps brachii, infraspinatus, and serratus anterior (SA) muscles were assessed with a surface electromyography device. Results of this study showed that elbow position changes resulted in higher LT ($p = 0.01$) and TB ($p = 0.001$) activation in HP exercises. In general, it was observed that an increase in BWB status was effective in increasing activation for the scapula and shoulder girdle muscles. The ratios of the UT muscle to the SA, LT, and MT muscles were less than 1 during side plank, bird dog, front reach, shoulder taps, and plank with shoulder external rotation exercises. High plank with toe touch exercise resulted in a marked increase in the UT activation. Therefore, it was concluded plank variations alter activation levels and activation ratios of shoulder and scapular muscles. Plank exercises may be used in shoulder rehabilitation and the progression in plank variations can be achieved by changing elbow position and BWB status based on individual requirements.

Key Words: exercise, rehabilitation, electromyography, shoulder joint

Introduction

Restoring neuromuscular control by addressing strength deficits and activation imbalances in the force couples is one of the main aims of shoulder rehabilitation (2). Previous research has shown that individuals experiencing shoulder pain often exhibit activation imbalance between force couples. Specifically, increased muscle activation of upper trapezius (UT) along with a decrease in muscular activation in the serratus anterior (SA), lower trapezius (LT), and middle trapezius (MT) muscles is frequently encountered (5,16,20). Given previous reports of activation imbalances in force couples in shoulders with pathologies, it is important to select exercises that promote optimal muscular activation ratios for improved neuromuscular control (5,16,20). Therefore, UT/SA, UT/LT, and UT/MT activation ratios are particularly important in shoulder rehabilitation. However, previous recommendations have been widely based on open kinetic chain exercise options (6,14,23).

Closed kinetic chain exercises are defined as exercises where the distal extremity is fixed to a stable or unstable floor and the body weight is carried on the distal segment (30). Closed kinetic chain exercises are considered advantageous for providing joint stability by providing co-contraction of the antagonist and agonist muscles around the joint (19). To restore functional stability in shoulder rehabilitation, it is recommended to use CKC exercises that carry body weight from the early period (19,33).

However, there are limited data about changes in the activation levels and relative activation ratios of the shoulder girdle muscles and scapular muscles during CKC exercises.

Plank exercises are frequently performed to strengthen the core region in rehabilitation programs (29). However, plank exercises are performed by carrying body weight on the upper extremity; therefore, this exercise places a load on the scapular muscles as well. During standard plank, the body weight is distributed over the 4 extremities in contact with the ground. Plank variations can be achieved by making changes in extremity positions or by adding movements. For example, reducing the number of weight-bearing extremities would result in varying levels of body-weight-bearing (BWB). In a study investigating CKC exercises, higher scapular muscle activation levels (UT, LT, and SA) were observed in the 3-pod plank exercise, in which the opposite hand touched the diagonal shoulder, than in the plank exercise in which the hands remained fixed on a balance board (25). It has been observed that increased body weight load on the shoulder and changes in stability also increase muscle activation levels (pectoralis major, anterior deltoid, infraspinatus [IS], SA, and latissimus dorsi) (11). However, it was seen that the increases in muscle activation levels were more in favor of UT as the difficulty level of the unstable platform used increased (1,2). Conversely, relative activation ratios of the scapular muscles were found to be lower in 3-pod plank exercises performed on 1 hand (25,26). Therefore, it is believed that while increasing the difficulty level of plank variations in shoulder rehabilitation, decreasing extremity support instead of using an unstable floor will be more effective in maintaining optimal scapular muscle activation ratios.

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Journal of Strength and Conditioning Research 00(00)/1–8

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During the low plank (LP) exercise, the elbows are flexed and fixed on the ground. In the high plank (HP) exercise, which is a variation of the LP exercise, the elbows are extended and the body weight is carried on the hands. It is believed that this change in the elbow position may provide changes in the muscle activation levels around the shoulder. There is limited information about the muscular activation levels seen in the variations of plank exercises performed by carrying body weight on the hand or elbow, and studies are needed to assist in optimal exercise selection. Investigating the scapula and shoulder muscle activation levels of different plank variations and optimal scapular muscle ratios will help clinicians in the selection of evidence-based exercises in developing rehabilitation programs.

The aim of this study was to investigate the activation levels and activation ratios of shoulder and scapular muscles during LP and HP variations, with varying levels of body weight support, including 4-point, 3-pod, and 2-point BWB statuses. It was hypothesized that there would be differences in the activation of scapular and shoulder muscles, as well as in activation ratios, between LP and HP variations, and as body weight support increased. The results of this study may provide valuable insights into scapular and shoulder girdle neuromuscular control and assist movement specialists in selecting and progressing exercise regimes for shoulder health.

Methods

Experimental Approach to the Problem

This cross-over study was conducted in a single session for each subject. Experimental sessions included the maximal voluntary isometric contraction (MVIC) tests and 10 plank variations. Maximal voluntary isometric contraction tests were conducted for the UT, MT, LT, SA, triceps brachii (TB), biceps brachii (BB), and IS muscles, and muscle activation levels were measured during plank variations. The experiments were randomly assigned and performed on the same day.

Subjects

Twenty-one healthy male individuals (mean $\pm$ SD, 26 ± 6.5 years of age and 24.4 ± 2.4 kg·m⁻² BMI) participated in this study. Sample size analysis was performed using the G*Power program, which predicted that at least 21 individuals were required to show an average 20% difference in muscle activation levels between movements, assuming a 30% SD, with a 5% type 1 error and 80% power. Individuals who were physically active were included in the study. They were required to do moderate-level or high-level regular exercise-like fitness or functional trainings at least 2 days a week. Individuals who were pain-free, aged between 18 and 45 years, had 5 points or above on the Tegner Activity Score (31), had less than 30 kg·m⁻² body mass index, had full active shoulder range of motion with no pain, and had no symptoms in screening tests including Hawkins-Kennedy, Neer, apprehension, and Jobe tests were included in this study (7). The exclusion criteria were as follows: history of upper-extremity injury or surgery and any musculoskeletal or systemic disease. Subjects were provided with detailed information before enrolling in the study, and written consent was obtained from all subjects. This study was performed with the approval of the Hacettepe University Institutional Review Board.

Procedures

This study was performed in the institutional movement laboratory. Before the surface electromyography (EMG) recordings,

subjects were evaluated for inclusion and exclusion criteria. Electrode placement was performed on the subjects who were eligible for the study. Then, MVIC tests and 10 plank variations were performed. Finally, the EMG data were analyzed.

Electromyography. Upper trapezius, MT, LT, SA, TB, BB, and IS muscle activation levels were measured using a surface EMG system (TeleMyo DTS system; Noraxon, Scottsdale, AZ) with 8 channels. Measurements were made only for the dominant shoulder. Electrode placement areas of the skin were first shaved and then cleaned with 70% isopropyl alcohol to minimize skin impedance (<10 k Ω). The sampling rate was set to 1,500 Hz, the gain was 1,000 (baseline noise <1 μ V root mean square), and the common-mode rejection ratio was >80 dB. Surface EMG for the noninvasive assessment of muscles (SENIAM) criteria were considered for electrode placements (10). Bipolar silver/silver chloride surface electrodes were used, with a distance of 2 cm between them. For the electrode placement of the UT, surface electrodes were placed in the middle of the seventh cervical vertebra and the acromion. For the MT, surface electrodes were placed in the middle of the third thoracic vertebra and the medial edge of the scapula. For the LT, surface electrodes were placed at 2 of 3 of the line between the trigonum spinae and the eighth thoracic vertebra. For the SA, surface electrodes were placed above the fourth to sixth ribs at the midaxillary line. For the IS, surface electrodes were placed 3 cm below the spina scapula. For the BB, surface electrodes were placed at 1 of 3 of the line between the fossa cubital and the medial acromion, starting at the fossa cubital. For the TB, surface electrodes were placed 2 fingers medial to the midpoint of the line between the olecranon and posterior crest of the acromion (Figure 1).

Maximal Voluntary Isometric Contraction Tests. Maximal voluntary isometric contractions and the exercises were performed in a mixed order to prevent order bias. The same investigator (E.N.C.) performed the MVIC test. Standardized verbal encouragements were given to promote maximal contraction during the measurements for each subject. Manual muscle testing positions were used as the MVIC test positions (1,3,24). Resistance was applied manually by the investigator. Practice trials were performed for the MVIC test positions before data collection. For the MVIC measurement of the UT, the shoulder was abducted at 90° while the subject was in the sitting position and the resistance was applied downward from the elbow (3). For the MVIC measurement of the MT, the resistance was applied downward from the elbow while the subject was in the prone position and the shoulder was in horizontal abduction and full external rotation (3). For the MVIC measurement of the LT, the arm was raised to the direction of the LT muscle fibers while the subject was in the prone position and downward resistance was applied over the elbow (3). For the MVIC measurement of the SA muscle, the shoulder was flexed 135° while the individual was in a sitting position and the resistance was applied downward from the elbow (3). For the MVIC measurement of the IS, while the elbow was flexed 90°, the shoulder was supported by a towel roll and abducted approximately 10°, and the resistance was applied from the wrist to the medial direction (24). For the MVIC measurement of the BB, the resistance was applied downward from the wrist while the subject was in a sitting position, the elbow was 90° flexed and the forearm was supinated (1). For the MVIC measurement of the TB, the elbow was flexed 90° and the forearm was in the neutral position while the subject was in the sitting position and the resistance was applied from the wrist upward (1). Three

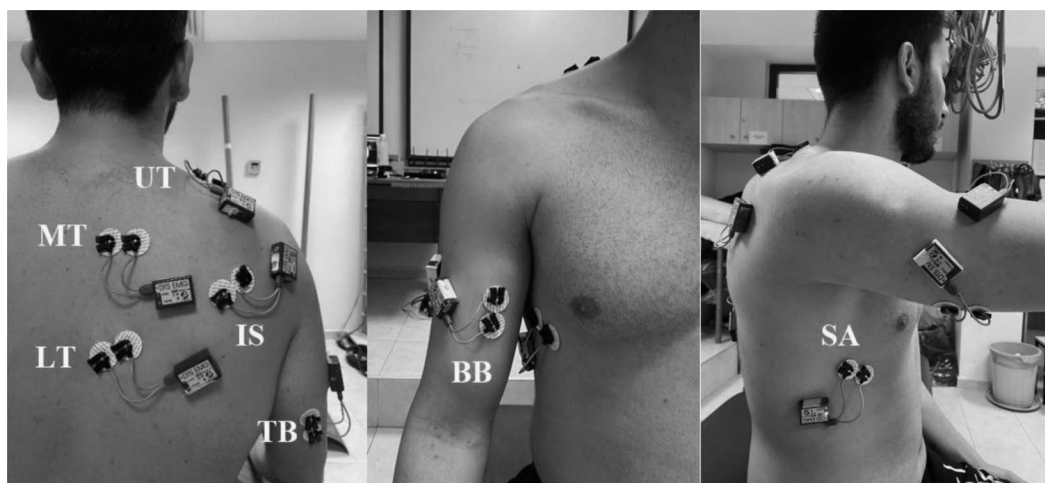


Figure 1. Electrode locations of the upper trapezius (UT), middle trapezius (MT), lower trapezius (LT), serratus anterior (SA), infraspinatus (IS), biceps brachii (BB), and triceps brachii (TB) muscles according to the SENIAM criteria.

test trials were performed in each MVIC position, and each trial took 5 seconds. Between trials, 30 seconds of rest periods were given. One-minute rest periods were given between each muscle testing.

Exercise Testing. Ten distinctive plank exercises were selected from the LP and HP variations performed with different BWB statuses: 4-point, 3-pod, and 2-point in a random order. Low plank exercises are the plank performed with elbows and toes on the floor, whereas HP exercises are the plank performed with hands and toes on the floor. Four-point plank variations are exercises where both upper extremity and both toes are in contact with the ground. Three-pod plank variations are exercises where 1 extremity is not in contact with the ground. Two-point plank variations are exercises where 1 upper and 1 lower extremity are not in contact with the ground. All exercises are presented in Table 1.

The subjects were instructed to maintain a standard plank position and get familiarized with exercises during the experiments (12). The standard plank position was explained to the subjects, along with instructions on how to maintain a neutral lumbopelvic and scapular position. Subjects experimented with each plank variation until they could perform it correctly. These procedures were repeated until the exercise positions were learned. After the individuals were ready to maintain the plank position while following specific instructions for each variation, the subjects were asked to hold it for 10 seconds and EMG data were recorded. Each exercise was repeated 3 times with 1-minute rest periods between each repetition. To reduce the effect of fatigue on the results, 2-minute rest periods were given between each exercise.

Electromyography Signal Processing. Noraxon myoRESEARCH (Noraxon TeleMyo DTS system) program was used for EMG signal processing (17). Electromyography signals were band-pass filtered (20 Hz) and smoothed with a 100-millisecond time reference using root mean square (8). For each exercise trial, 10-second recordings were made. The mean muscle activation levels of the middle 5 seconds of the 10-second data collection period were recorded. Then, the mean of the 3 trials was normalized to the maximum activation level recorded in MVIC tests. Normalized EMG signals are expressed as %MVIC. %MVIC values of the SA, LT, MT, UT, IS, BB, and TB muscles and the BB/TB, UT/

SA, UT/LT, and UT/MT activation ratios were calculated for each exercise. Maximal voluntary isometric contraction levels between 0 and 20% were considered low intensity, 21–40% moderate, 41–60% high, and 60% and above very high muscle activation (9). UT/SA, UT/LT, and UT/MT ratios less than 1 were considered optimal based on recommendations by Schory et al. (27).

Statistical Analyses

SPSS program (version 21.9; IBM Corp, IL) was used for statistical analysis of this study. The normality of data distribution was determined using the Kolmogorov-Smirnov test. Student's *t* test was applied to investigate the individual muscle activation levels and relative activation ratios between the LP and HP exercises. To assess muscle activation levels among the 4-point, 3-pod, and 2-point BWB statuses for LP and HP variations, repeated-measures analysis of variance (ANOVA) tests were performed for each tested muscle. Repeated-measures ANOVA tests were also performed to analyze relative muscular activity ratios (BB/TB, UT/SA, UT/LT, and UT/MT). Bonferroni correction was used for post hoc pairwise comparisons. Statistical significance level was set as $\alpha < 0.05$.

Results

Plank variations mostly revealed low to moderate activation for the scapular and shoulder muscles. Serratus anterior revealed a high muscular activation during the 2-point LP-BD, and moderate activation levels were recorded for the other 9 plank variations. In addition, moderate activation levels were observed for LT during HP-SP; for IS during LP-ER, LP-FR, LP-BD, HP, HP+, HP-ST, and HP-SP; and, for TB during LP-FR, LP-BD, HP-ST, HP-TT, and HP-SB (Figure 2). Significant differences were noted between plank variations in their ability to achieve scapular and shoulder muscle activation levels, as well as in relative activation ratios.

Muscular Activation Levels

Analysis of variance revealed a significant effect of the elbow position on LT and TB muscle activation when LP and HP exercises were compared. Lower trapezius and TB muscles showed

Table 1
Matrix of the exercises.

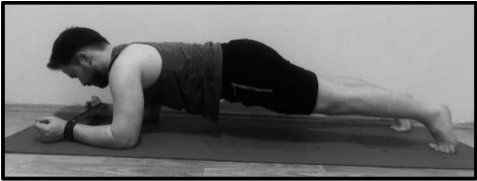
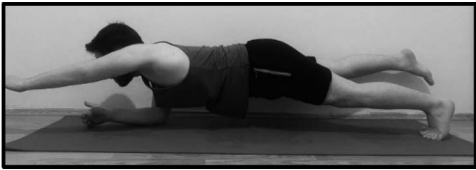
	Low plank variations	High plank variations
4-point exercises	 Low plank (LP): Elbows and toes are on the floor and shoulder-width apart. The whole body was held in a straight position at neutral angles, with the scapula in the midretraction position.  Low plank w/isometric shoulder external rotation (LP-ER): Elbows and toes are on the floor and shoulder-width apart. Resistance to external rotation was provided with a blue elastic band to be passed around the elbow. The whole body was kept in a straight position at neutral angles, with the scapula in the midretraction position.	 High plank (HP): Hands and toes are on the floor and shoulder-width apart. The whole body was held in a straight position at neutral angles, with the scapula in the midretraction position.  High plank plus (HP+): Hands and toes were positioned on the floor and shoulder-width apart. It was started in a straight position with the whole body at neutral angles. Maximal protraction of the scapula was requested by controlling the thoracic kyphosis.  Bear plank (HP-BP): Hips and knees were bent at 90° flexion, and while the hands were below the shoulder level, the knees were lifted up to 5 cm above the heel level.
3-pod exercises	 Low plank w/front reach (LP-FR): Elbows and toes are on the floor and shoulder-width apart. The whole body was started in a straight position at neutral angles. The subject was asked to reach forward with the contralateral arm without changing the body position.	 High plank w/shoulder taps (HP-ST): Hands and toes were started on the floor and shoulder-width apart. While the whole body was in a straight position at neutral angles, the opposite hand was raised so that the dominant side touched the shoulder.  High plank w/toe touch (HP-TT): Hands and toes were started on the floor and shoulder-width apart. With hip flexion, the contralateral hand was extended toward the cross-side foot.

Table 1**Matrix of the exercises.** (Continued)

	Low plank variations	High plank variations
2-point exercises	 Low plank w/bird dog (LP-BD): Elbows and toes were started on the floor and shoulder-width apart. While the nondominant arm was extended forward, the cross-side leg was lifted from the ground without deteriorating the body position.	 High side plank (HP-SP): Body is lifted into the air with the dominant side hand below shoulder level, with the whole body in a straight line from the top of the head to the feet.

significantly higher activation in HP exercises than in LP exercises ($p = 0.01$, mean difference (MD) = 3.2 for LT and $p = 0.001$, MD = 5.5 for TB). There were no significant statistical differences between HP and LP exercises for UT, MT, SA, IS, and BB muscle activation levels.

Body-weight-bearing status resulted in a significant difference in muscle activation levels during LP variations. Upper trapezius, MT, IS, BB, and TB activation levels were higher in the 3-pod LP-FR exercise and the 2-point LP-BD exercise compared with the 4-point LP and LP-ER exercises (Figure 2). Serratus anterior

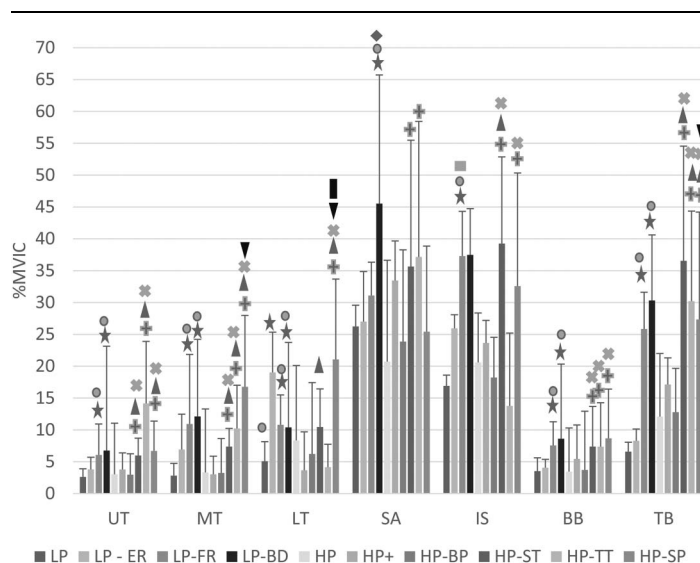


Figure 2. Activation levels of scapula and shoulder girdle muscles during LP and HP variations. MVIC, maximum voluntary isometric contraction; UT, upper trapezius; MT, middle trapezius; LT, lower trapezius; SA, serratus anterior; IS, infraspinatus; BB, biceps brachii; TB, triceps brachii; LP, low plank; LP-ER, low plank w/isometric shoulder external rotation; LP-FR, low plank w/front reach; LP-BD, low plank w/bird dog; HP, high plank; HP+, high plank plus; HP-BP, high plank w/bear plank; HP-ST, high plank w/shoulder taps; HP-TT, high plank w/toe touch; HP-SP, high side plank. Results of pairwise comparisons marked when statistical difference was found when compared with LP (★); results of pairwise comparisons marked when statistical difference was found when compared with LP-ER (●); results of pairwise comparisons marked when statistical difference was found when compared with LP-FR (◆); results of pairwise comparisons marked when statistical difference was found when compared with LP-BD (■); results of pairwise comparisons marked when statistical difference was found when compared with HP (+); results of pairwise comparisons marked when statistical difference was found when compared with HP+ (▲); results of pairwise comparisons marked when statistical difference was found when compared with HP-BP (✕); results of pairwise comparisons marked when statistical difference was found when compared with HP-ST (▼); results of pairwise comparisons marked when statistical difference was found when compared with HP-TT (■).

Table 2**Activation ratios of selected upper-extremity muscles during LP variations.*†**

	LP	LP-ER	LP-FR	LP-BD	<i>p</i> ‡
	Mean ± SD	Mean ± SD	Mean ± SD	Mean ± SD	
UT/MT	1.19 ± 0.78	0.79 ± 0.56	0.72 ± 0.44	0.72 ± 0.44	<0.001§
UT/LT	0.77 ± 0.49	0.26 ± 0.19	0.62 ± 0.28	0.78 ± 0.51	<0.001§
UT/SA	0.12 ± 0.09	0.17 ± 0.13	0.25 ± 0.22	0.19 ± 0.14	0.001§
BB/TB	0.63 ± 0.43	0.63 ± 0.68	0.3 ± 0.18	0.31 ± 0.2	0.005

*UT, upper trapezius; MT, middle trapezius; LT, lower trapezius; SA, serratus anterior; IS, infraspinatus; BB, biceps brachii; TB, triceps brachii; LP, low plank; LP-ER, low plank w/isometric shoulder external rotation; LP-FR, low plank w/front reach; LP-BD, low plank w/bird dog.

†Bold type of results shows optimal muscular activation ratios based on Schory et al. (27).

‡Shows repeated-measures ANOVA results.

§Shows statistical differences.

showed the highest activation in the 2-point LP-BD (mean ± SD; 45.53 ± 20.2) compared with LP, LP-ER, and LP-FR ($p = 0.002$, MD = 19.2 for LP; $p = 0.002$, MD = 18.5 for LP-ER; and $p = 0.002$, MD = 14.4 for LP-FR). Lower trapezius had the highest activation during LP-ER (19.02 ± 10.92), compared with LP, LP-FR, and LP-BD ($p < 0.001$, MD = 13.9 for LP; $p = 0.009$, MD = 8.2 for LP-FR; and $p = 0.007$, MD = 8.6 for LP-BD). In addition, LT had the lowest activation during LP (5.08 ± 4.86) compared with LP-ER, LP-FR, and LP-BD ($p < 0.001$, MD = 5.7 for LP-FR and $p = 0.001$, MD = 5.3 for LP-BD).

Body-weight-bearing status resulted in a significant difference in muscle activation levels during HP variations. Upper trapezius, MT, BB, and TB activation levels were higher in the 3-pod HP-ST and HP-TT and 2-point HP-SP exercises compared with 4-point HP, HP+, and HP-BP exercises. Lower trapezius had the highest activation during HP-SP (21.06 ± 12.63) compared with HP, HP+, HP-BP, HP-ST, and HP-TT ($p = 0.004$, MD = 12.7 for HP; $p < 0.001$, MD = 17.3 for HP+; $p < 0.001$, MD = 14.8 for HP-BP; $p = 0.01$, MD = 10.6 for HP-ST; and $p < 0.001$, MD = 16.9 for HP-TT). Serratus anterior showed higher activation in the HP+, HP-ST, and HP-TT exercises compared with HP, HP-BP, and HP-SP exercises. Infraspinatus had the highest activation during HP-ST (39.25 ± 13.62) compared with HP, HP+, HP-BP, and HP-TT exercises ($p < 0.001$, MD = 18.6 for HP; $p < 0.001$, MD = 15.5 for HP+; $p < 0.001$, MD = 21 for HP-BP; and $p < 0.001$, MD = 25.4 for HP-TT).

Relative Muscular Activation Ratios

Relative muscle activation ratios during LP and HP variations are presented in Tables 2 and 3. Analysis of variance revealed no significant effect of the elbow position on BB/TB, UT/SA, UT/LT,

and UT/MT activation ratios when the LP and HP exercises were compared ($p > 0.05$).

The highest UT/MT ratio was observed during the LP compared with the LP-ER, LP-FR, and LP-BD exercises ($p < 0.001$, $p = 0.003$, and $p = 0.01$, respectively). The lowest UT/LT ratio was observed during the LP-ER (0.26 ± 0.19) compared with the LP, LP-FR, and LP-BD exercises ($p < 0.001$, $p < 0.001$, and $p = 0.001$, respectively). The lowest UT/SA ratio was observed during the LP exercises (0.12 ± 0.09) compared with the LP-FR and LP-BD exercises ($p = 0.005$ and $p = 0.01$, respectively). Lower ratios of BB/TB were observed during the 3-pod LP-FR exercise and the 2-point LP-BD exercise compared with the 4-point LP and LP-ER exercises.

The lowest UT/MT ratio was observed during the HP-SP (0.54 ± 0.51) compared with the HP, HP+, HP-BP, HP-ST, and HP-TT exercises ($p < 0.001$, $p < 0.001$, $p < 0.001$, $p = 0.04$, and $p = 0.02$, respectively). The lowest UT/LT ratio was observed during the HP-SP (0.54 ± 0.51) compared with the HP+, HP-ST, and HP-TT exercises ($p = 0.001$, $p = 0.005$, and $p < 0.001$, respectively). The UT/LT ratio during the HP-TT exercises (4.56 ± 3.49) was statistically higher than those of the HP, HP+, HP-BP, HP-ST, and HP-SP exercises ($p < 0.001$, $p = 0.02$, $p = 0.002$, $p = 0.001$, and $p < 0.001$, respectively). The lowest BB/TB ratio was observed during the HP-ST (0.21 ± 0.15) compared with the HP+ and HP-SP exercises ($p = 0.04$ and $p = 0.008$, respectively).

Discussion

This study investigated the activation levels and activation ratios of shoulder and scapular muscles during LP and HP variations, with varying levels of body weight support, including 4-point, 3-pod, and 2-point BWB statuses. When LP and HP exercises were compared, altering elbow position resulted in differences in LT and TB muscle activation levels; however, there were no differences in relative muscle activation ratios. In general, BWB status in plank variations had an effect on muscle activation levels and relative muscle activation ratios.

This study is important because it is the first study to compare the activation levels of the scapula and shoulder muscles during LP and HP exercises and 4-point, 3-pod, and 2-point variations of these exercises. When the LP and HP exercises were compared at the same BWB status, it was seen that LT and TB muscle activation levels were higher in the HP exercise. Decreased support surface in the HP exercise may have caused more activation of the LT muscle to provide scapular stabilization. In clinical situations such as scapular dyskinesis where LT activation is lower than other scapular muscles, the HP exercise may be a more appropriate option for exercise

Table 3**Activation ratios of selected upper-extremity muscles during HP variations.*†**

	HP	HP+	HP-BP	HP-ST	HP-TT	HP-SP	<i>p</i> ‡
	Mean ± SD	Mean ± SD	Mean ± SD	Mean ± SD	Mean ± SD	Mean ± SD	
UT/MT	1.14 ± 0.74	1.39 ± 0.79	1.16 ± 0.78	0.9 ± 0.52	2.21 ± 2.2	0.54 ± 0.51	0.006§
UT/LT	0.52 ± 0.43	1.59 ± 1.14	0.87 ± 0.8	0.76 ± 0.48	4.56 ± 3.49	0.4 ± 0.32	<0.001§
UT/SA	0.16 ± 0.09	0.13 ± 0.13	0.14 ± 0.09	0.21 ± 0.15	0.54 ± 0.53	0.32 ± 0.29	0.002§
BB/TB	0.32 ± 0.28	0.36 ± 0.32	0.32 ± 0.31	0.21 ± 0.15	0.26 ± 0.2	0.32 ± 0.24	0.02§

*UT, upper trapezius; MT, middle trapezius; LT, lower trapezius; SA, serratus anterior; IS, infraspinatus; BB, biceps brachii; TB, triceps brachii; HP, high plank; HP+, high plank plus; HP-BP, bear plank; HP-ST, high plank w/shoulder taps; HP-TT, high plank w/toe touch; HP-SP, high side plank.

†Bold type of results shows optimal muscular activation ratios based on Schory et al. (27).

‡Shows repeated-measures ANOVA results.

§Shows statistical differences.

selection. However, there were no differences in relative muscle activation ratios between the HP and LP exercises. Therefore, the results of our study determined that the LP and HP exercises were not superior to each other in preserving the relative muscle activation ratios of the scapular muscles and the muscles around the elbow.

There are previous reports on 4-point and 3-pod variations of the plank exercise (11,22,25,34). In previous studies, higher UT, MT, LT, SA, and IS muscle activation levels have been observed in the 3-pod plank and push-up variations compared with the 4-point exercises (11,21,25,32). Similar to previous studies, in our study, it was observed that the BB, TB, UT, and MT muscle activation levels increased in the 3-pod and 2-point plank variations, where the body weight load carried on the extremities increased. Compared with the 4-point variations, higher IS activation levels were seen in the 3-pod and 2-point LP variations. In addition, the highest IS activation in the HP variations was seen in the 3-pod HP-ST exercise. Among the HP variations, the LT muscle showed the highest activation in the 2-point HP-SP exercise, whereas among the LP variations, it showed the highest activation in the LP-ER exercise with resistance to external rotation. Although the LP-ER exercise was performed with 4-point, adding resistance to external rotation as an additional movement increased muscle activation. Similarly, in previous studies, it was observed that exercises performed together with external rotation provided more LT activation (6,35). Among the LP variations, the SA muscle showed the highest activation in the 2-point LP-BD exercise, whereas among the HP variations, it showed the highest activation in the 3-pod exercises and the HP+ exercise with scapular protraction. Similarly, in previous studies, it was determined that there was an increase in SA activation in the plus phase of the push-up plus exercise (13). The results of our study provide valuable information for developing progressive exercise programs for the scapula and shoulder girdle muscles. In exercise programs that start with 4-support plank exercises, the exercise intensity can be increased by decreasing the support level or adding movements such as resisted shoulder external rotation.

To maintain scapular muscle balance in shoulder rehabilitation, a ratio of UT muscle activation to MT, LT, and SA muscle activation levels less than 1 was considered optimal during exercises (27). In previous studies, it has been observed that the ratios of the scapular muscles were optimal in the 3-pod plank exercise (22). In this study, the UT/SA and UT/LT ratios in all LP variations with different support levels and the UT/MT ratio in all variations except for the 4-point LP exercise were optimal. In addition, among the HP variations, the UT/SA, UT/LT, and UT/MT ratios were optimal in the 2-point HP-SP and 3-pod HP-ST exercises. When the results are examined, the 3-pod and 2-point exercises are more effective in maintaining scapular ratios, especially for HP variations. However, the UT/LT and UT/MT ratios were quite high (>1) due to the excessive increase in UT activation in the HP-TT exercise examined in this study. It has also been shown in previous studies that an increase in shoulder flexion angle in the push-up position causes an increase in UT activation (18). For this reason, it is not recommended to use exercises such as HP-TT in shoulder rehabilitation in which the shoulder flexion angle exceeds 90° .

Previously, it has been observed that exercises performed with external rotation provide more LT and MT activation, along with lower UT/LT and UT/MT activation ratios (6,35). Similar to these studies, in our study, the LP-ER exercise performed by adding elastic band resistance against external

rotation around the wrist provided an increase in MT and LT activation. Among the 4 assisted exercises, the exercise that provided the highest MT and LT activation without excessive increase in UT activation was the LP-ER exercise. This exercise also provided the lowest UT/LT and UT/MT ratios of all exercises. Therefore, the LP-ER exercise can be used in clinics to activate the scapular muscles when excessive activation of the UT muscle is not desired.

In pathologies associated with the BB muscle, the load on the muscle should be minimal in the early period of rehabilitation programs (4). In rehabilitation programs involving the elbow joint, it is important to include exercises that provide TB muscle activation in the program for functional joint stabilization. During the exercises tested in our study, BB and TB activation levels were at low intensity ($<20\%$) in all 4-point exercises. As the support level decreased in the plank exercises, TB activation increased to moderate (20–40%) without increasing BB activation much. For this reason, the BB/TB ratio was lower in the 3-pod and 2-point exercises compared with the 4-point exercises. The results of the study show that, due to the low activation of the BB muscle, all plank variations used in this study can be used in rehabilitation programs in BB-related pathologies, for example, after SLAP surgery, and 3-pod and 2-point exercises can be used to provide more TB activation. However, there is a need for further studies involving the population who have undergone surgery.

The first limitation of our study is that the findings of this study represents only physically active, healthy male subjects, and female subjects or patients with any shoulder problems were not studied. In addition, the standard resistance of the color-coded elastic band used in the LP-ER exercise was not individualized. However, the decision of the color of elastic band was based on a preliminary study (unpublished data) to use the difficulty level as 6–8 on the OMNI scale. Finally, owing to the high number of exercises included in the test protocol and the challenging nature of the exercises, there was a risk of fatigue in individuals. However, to minimize the effect of fatigue, standard rest breaks were given between repetitions and all exercises were performed in a mixed order.

Practical Applications

Our findings support the idea that plank exercises can be used effectively in shoulder rehabilitation. LP-ER exercises, which provide scapular muscle activation without excessive increase in UT activation, can be recommended in individuals with significantly increased activation in the UT muscle and/or rotator cuff problems. In addition, it was observed that the addition of the elastic band against the external rotation in the LP exercise and the decrease of the support level increased the IS activation. Based on these results, it was determined that in clinical situations where rotator cuff activation is required, a progressive increase in exercise intensity can be achieved by increasing the body weight on the upper extremity or by providing resistance in the direction of external rotation. Owing to the increase in UT activation, care should be taken in the use of exercises with increased shoulder flexion angle, such as HP-TT exercises, in patients with shoulder problems. Owing to the low BB muscle activation in plank variations, it was concluded that all the exercises examined in this study can be used in BB-related problems, for example, in the early period of rehabilitation where BB activation should be restricted after certain surgeries.

Acknowledgments

ClinicalTrials.gov Identifier: NCT05281900. Ethical approval: Hacettepe University Institutional Review Board approved the protocol for this study (GO 21/979). Research laboratory: Hacettepe University, Faculty of Physical Therapy and Rehabilitation, Ankara, Turkey. The authors have no conflicts of interest to disclose. Authors' contribution: E. N. Can performed data acquisition and analysis and drafted the manuscript; E. Turgut has made substantial contributions to conception and design and revising the manuscript critically for important intellectual content; G. Harput was involved in data acquisition and analysis and made substantial contributions to conception and design and interpretation of data. All authors read and approved the final manuscript.

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